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PROBLEM STATEMENT:

OpinRank Hotel Reviews Analysis and Data Cleaning. You are provided with the world cup dataset containing information about hotel ratings, pricing, and reviews across major cities. Your task is to write Python code to answer the following questions based on the dataset. For each question perform necessary data cleaning transformation and calculations as required.

DATASET OVERVIEW:

Review_ID	Hotel_Name	City	Rating	Reviews_Count	Price_Per_Night	Cleanliness	Service	Location
1	Grand Hyatt	Dubai	4.5	1200	320	4.6	4.4	4.7
2	Marriott	London	4.2	980	280	4.3	4.1	4.5
3	Hilton	Paris	4.8	1450	350	4.7	4.9	4.6
4	Ritz Carlton	New York	4.9	870	480	4.9	4.8	4.9
5	Sheraton	Tokyo	3.9	630	260	4.0	3.8	4.4
6	Holiday Inn	Dubai	3.5	540	150	3.6	3.4	3.8

Using Pandas:

1. Display first 5 rows.
2. Display last 5 rows.
3. Find shape of the dataset.
4. Get the summary of dataset.
5. Get descriptive statistics.
6. Check missing values.
7. Create Value_Score column ($\text{Rating} / \text{Price_Per_Night} * 100$).
8. Create Avg_Score column (mean of Cleanliness, Service, Location scores).
9. Find hotel count by city.
10. Find year-wise average rating.

Using Numpy:

1. Find total reviews count in dataset.
2. Find average rating.
3. Find highest price per night.
4. Find lowest price per night.
5. Find median rating.
6. Find standard deviation of ratings.
7. Find index of hotel with highest reviews count.
8. Sort ratings in ascending order.
9. Find correlation between rating and price.
10. Find mean service score.

PYTHON CODE:

Dataset Creation & Library Imports:

```
import numpy as np
import pandas as pd

data = {
    "Review_ID": list(range(1, 16)),
    "Hotel_Name": [
        "Grand Hyatt", "Marriott", "Hilton", "Ritz Carlton", "Sheraton",
        "Holiday Inn", "Best Western", "Radisson", "Novotel", "Ibis",
        "Four Seasons", "Crown Plaza", "Westin", "InterContinental", "Sofitel"],
    "City": [
        "Dubai", "London", "Paris", "New York", "Tokyo",
        "Dubai", "Paris", "London", "Tokyo", "New York",
        "Dubai", "London", "Paris", "Tokyo", "New York"],
    "Rating": [4.5, 4.2, 4.8, 4.9, 3.9, 3.5, 3.2, 4.0, 4.3, 3.7, 5.0, 4.1, 4.6, 4.4, 4.7],
    "Reviews Count": [1200, 980, 1450, 870, 630, 540, 320, 760, 1100, 480, 1600, 900, 1300, 820, 1050],
    "Price_Per_Night": [320, 280, 350, 480, 260, 150, 110, 200, 270, 130, 550, 230, 310, 290, 370],
    "Cleanliness_Score": [4.6, 4.3, 4.7, 4.9, 4.0, 3.6, 3.3, 4.1, 4.4, 3.8, 5.0, 4.2, 4.5, 4.3, 4.8],
    "Service_Score": [4.4, 4.1, 4.9, 4.8, 3.8, 3.4, 3.1, 3.9, 4.2, 3.6, 5.0, 4.0, 4.7, 4.5, 4.6],
    "Location_Score": [4.7, 4.5, 4.6, 4.9, 4.4, 3.8, 3.5, 4.3, 4.5, 4.0, 4.8, 4.4, 4.7, 4.6, 4.5],
    "Year": [2021, 2021, 2022, 2022, 2021, 2023, 2023, 2022, 2023, 2023, 2021, 2022, 2023, 2022, 2021],
    "Recommend": ["Yes", "Yes", "Yes", "Yes", "No", "No", "No", "Yes", "Yes", "No",
                  "Yes", "Yes", "Yes", "Yes", "Yes"]
}

df = pd.DataFrame(data)
df["Value_Score"] = round(df["Rating"] / df["Price_Per_Night"] * 100, 2)
df["Avg_Score"] = round(
```

```
(df["Cleanliness_Score"] + df["Service_Score"] + df["Location_Score"]) / 3, 2)
```

Pandas Operations (1 – 10):

```
# 1. First 5 Rows
print('1. First Five Rows')
print(df.head())

# 2. Last 5 Rows
print('2. Last Five Rows')
print(df.tail())

# 3. Shape
print('3. Shape of Dataset')
print(df.shape)

# 4. Dataset Info
print('4. Dataset Info')
print(df.info())

# 5. Descriptive Statistics
print('5. Statistics')
print(df.describe())

# 6. Missing Values
print('6. Missing Values')
print(df.isnull().sum())

# 7. Value_Score column
print('7. Value Score Column')
print(df[["Hotel_Name", "Value_Score"]])

# 8. Avg_Score column
print('8. Average Score Column')
print(df[["Hotel_Name", "Avg_Score"]])

# 9. Hotels by City
print('9. Hotels by City')
print(df["City"].value_counts())

# 10. Year-wise Average Rating
print('10. Year-wise Average Rating')
```

```
print(df.groupby("Year")["Rating"].mean())
```

NumPy Operations (1 – 10):

```
ratings = np.array(df['Rating'])
prices = np.array(df['Price_Per_Night'])
reviews = np.array(df['Reviews_Count'])
service = np.array(df['Service_Score'])

# NumPy 1 - Total Reviews
print('NumPy 1 Total Reviews Count:', np.sum(reviews))

# NumPy 2 - Average Rating
print('NumPy 2 Average Rating:', round(np.mean(ratings), 4))

# NumPy 3 - Highest Price
print('NumPy 3 Highest Price/Night:', np.max(prices))

# NumPy 4 - Lowest Price
print('NumPy 4 Lowest Price/Night:', np.min(prices))

# NumPy 5 - Median Rating
print('NumPy 5 Median Rating:', np.median(ratings))

# NumPy 6 - Std Dev
print('NumPy 6 Std Dev of Ratings:', round(np.std(ratings), 4))

# NumPy 7 - Argmax Reviews
print('NumPy 7 Index Highest Reviews:', np.argmax(reviews))

# NumPy 8 - Sorted Ratings
print('NumPy 8 Sorted Ratings:', np.sort(ratings))

# NumPy 9 - Correlation
print('NumPy 9 Correlation Rating vs Price:')
print(np.corrcoef(ratings, prices))

# NumPy 10 - Mean Service
```

```
print('NumPy 10 Mean Service Score:', round(np.mean(service), 4))
```

PANDAS OUTPUT:

1. First Five Rows

OUTPUT

Review_ID	Hotel_Name	City	Rating	Reviews_Count	Price_Per_Night	
Cleanliness_Score	Service_Score	Location_Score	Year	Recommend	Value_Score	Avg_Score
0 4.6	1 4.4	Grand Hyatt	Dubai	4.5	1200	320
1 4.3	2 4.1	Marriott	London	4.2	980	280
2 4.7	3 4.9	Hilton	Paris	4.8	1450	350
3 4.9	4 4.8	Ritz Carlton	New York	4.9	870	480
4 4.0	5 3.8	Sheraton	Tokyo	3.9	630	260
				No	1.50	4.07
[5 rows x 13 columns]						

2. Last Five Rows

OUTPUT

Review_ID		Hotel_Name		City	Rating	Reviews_Count	Price_Per_Night	
Cleanliness_Score	Service_Score	Location_Score		Year	Recommend	Value_Score	Avg_Score	
10	11	Four Seasons		Dubai	5.0	1600	550	
5.0	5.0	4.8	2021	Yes	0.91	4.93		
11	12	Crown Plaza		London	4.1	900	230	
4.2	4.0	4.4	2022	Yes	1.78	4.20		
12	13	Westin		Paris	4.6	1300	310	
4.5	4.7	4.7	2023	Yes	1.48	4.63		
13	14	InterContinental		Tokyo	4.4	820	290	
4.3	4.5	4.6	2022	Yes	1.52	4.47		
14	15	Sofitel		New York	4.7	1050	370	
4.8	4.6	4.5	2021	Yes	1.27	4.63		
[5 rows x 13 columns]								

3. Shape of Dataset

OUTPUT

 $(15, 13)$

4. Dataset Info

OUTPUT

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Review_ID             15 non-null    int64
1   Hotel_Name            15 non-null    str
2   City                  15 non-null    str
3   Rating                15 non-null    float64
4   Reviews Count         15 non-null    int64
```

```

5   Price_Per_Night    15 non-null    int64
6   Cleanliness_Score  15 non-null    float64
7   Service_Score      15 non-null    float64
8   Location_Score     15 non-null    float64
9   Year               15 non-null    int64
10  Recommend          15 non-null    str
11  Value_Score        15 non-null    float64
12  Avg_Score          15 non-null    float64
dtypes: float64(6), int64(4), str(3)
memory usage: 1.7 KB
None

```

5. Descriptive Statistics

OUTPUT

```

      Review_ID    Rating  Reviews_Count  Price_Per_Night  Cleanliness_Score
Service_Score  Location_Score      Year  Value_Score  Avg_Score
count  15.000000  15.000000    15.000000    15.000000    15.000000
15.000000    15.000000    15.000000    15.000000    15.000000
mean    8.000000  4.253333    933.333333    286.666667    4.300000
4.200000    4.413333    2022.000000    1.696000    4.304667
std    4.472136  0.526263    362.228552    121.164980    0.484031
0.569461    0.381476    0.845154    0.592631    0.468567
min    1.000000  3.200000    320.000000    110.000000    3.300000
3.100000    3.500000    2021.000000    0.910000    3.300000
25%    4.500000  3.950000    695.000000    215.000000    4.050000
3.850000    4.350000    2021.000000    1.390000    4.085000
50%    8.000000  4.300000    900.000000    280.000000    4.300000
4.200000    4.500000    2022.000000    1.500000    4.370000
75%   11.500000  4.650000   1150.000000    335.000000    4.650000
4.650000    4.650000    2023.000000    1.890000    4.630000
max    15.000000  5.000000   1600.000000    550.000000    5.000000
5.000000    4.900000    2023.000000    2.910000    4.930000
[8 rows x 10 columns]

```

6. Missing Values

OUTPUT

```

Review_ID      0
Hotel_Name     0
City           0
Rating         0
Reviews_Count  0
Price_Per_Night 0
Cleanliness_Score 0
Service_Score  0
Location_Score 0
Year           0
Recommend      0
Value_Score    0
Avg_Score      0
dtype: int64

```

7. Value_Score Column (Rating / Price * 100)

OUTPUT		
	Hotel_Name	Value_Score
0	Grand Hyatt	1.41
1	Marriott	1.50
2	Hilton	1.37
3	Ritz Carlton	1.02
4	Sheraton	1.50
5	Holiday Inn	2.33
6	Best Western	2.91
7	Radisson	2.00
8	Novotel	1.59
9	Ibis	2.85
10	Four Seasons	0.91
11	Crown Plaza	1.78
12	Westin	1.48
13	InterContinental	1.52
14	Sofitel	1.27

8. Avg_Score Column (Mean of Cleanliness, Service, Location)

OUTPUT		
	Hotel_Name	Avg_Score
0	Grand Hyatt	4.57
1	Marriott	4.30
2	Hilton	4.73
3	Ritz Carlton	4.87
4	Sheraton	4.07
5	Holiday Inn	3.60
6	Best Western	3.30
7	Radisson	4.10
8	Novotel	4.37
9	Ibis	3.80
10	Four Seasons	4.93
11	Crown Plaza	4.20
12	Westin	4.63
13	InterContinental	4.47
14	Sofitel	4.63

9. Hotel Count by City

OUTPUT		
City		
Dubai	3	
London	3	
Paris	3	
New York	3	
Tokyo	3	
Name: count, dtype: int64		

10. Year-wise Average Rating

OUTPUT		
--------	--	--

```
Year
2021    4.46
2022    4.44
2023    3.86
Name: Rating, dtype: float64
```

NUMPY OUTPUT:

NumPy 1. Total Reviews Count

OUTPUT

```
np.sum(reviews) → 14000
```

NumPy 2. Average Rating

OUTPUT

```
np.mean(ratings) → 4.2533
```

NumPy 3. Highest Price Per Night

OUTPUT

```
np.max(prices) → 550
```

NumPy 4. Lowest Price Per Night

OUTPUT

```
np.min(prices) → 110
```

NumPy 5. Median Rating

OUTPUT

```
np.median(ratings) → 4.3
```

NumPy 6. Standard Deviation of Ratings

OUTPUT

```
np.std(ratings) → 0.5084
```

NumPy 7. Index of Hotel with Highest Reviews Count

OUTPUT

```
np.argmax(reviews) → 10 (Four Seasons, 1600 reviews)
```

NumPy 8. Sorted Ratings (Ascending)

OUTPUT

```
np.sort(ratings) → [3.2 3.5 3.7 3.9 4. 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5. ]
```

NumPy 9. Correlation between Rating and Price

OUTPUT

```
np.corrcoef(ratings, prices) →  
[[1.          0.92490483]  
 [0.92490483 1.          ]]
```

Interpretation: Strong positive correlation (0.9249) between rating and price.

NumPy 10. Mean Service Score

OUTPUT

```
np.mean(service) → 4.2
```